Given:

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x	y
2	3
3	5
5	7
7	10
9	12

1. $X = \begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 1 & 5 \\ 1 & 7 \\ 1 & 9 \end{bmatrix} \quad y = \begin{bmatrix} 3 \\ 5 \\ 7 \\ 10 \\ 12 \end{bmatrix}$

2. $X^T = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 2 & 3 & 5 & 7 & 9 \end{bmatrix}$

3. $X^T X = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 2 & 3 & 5 & 7 & 9 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 1 & 5 \\ 1 & 7 \\ 1 & 9 \end{bmatrix} = \begin{bmatrix} 5 & 26 \\ 26 & 168 \end{bmatrix}$

4. $X^T y = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 2 & 3 & 5 & 7 & 9 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \\ 7 \\ 10 \\ 12 \end{bmatrix} = \begin{bmatrix} 37 \\ 234 \end{bmatrix}$

5. $(X^T X)^{-1} = \frac{1}{(5)(168) - 26^2} \begin{bmatrix} 168 & -26 \\ -26 & 5 \end{bmatrix} = \frac{1}{164} \begin{bmatrix} 168 & -26 \\ -26 & 5 \end{bmatrix}$

6. $\hat{\beta} = \frac{1}{164} \begin{bmatrix} 168 & -26 \\ -26 & 5 \end{bmatrix} \begin{bmatrix} 37 \\ 234 \end{bmatrix} = \frac{1}{164} \cdot \begin{bmatrix} 132 \\ 208 \end{bmatrix} = \begin{bmatrix} .805 \\ 1.268 \end{bmatrix} = \begin{bmatrix} B_0 \\ B_1 \end{bmatrix}$

$\hat{y} = B_0 + B_1 x$ so $\hat{y} = .805 + 1.268x$

Part 2:

1. $\bar{x} = \frac{1}{5}\sum x_i = 5.2$, $\bar{y} = \frac{1}{5}\sum y_i = 7.4$

2. $\sum (x_i - \bar{x})(y_i - \bar{y}) = 41.6$, $\sum (x_i - \bar{x})^2 = 32.8$

3. $B_1 = \frac{41.6}{32.8} = 1.268$, $B_0 = 7.4 - \left(\frac{41.6}{32.8}\right)(5.2) = .805$

$\hat{y} = .805 + 1.268x$

Comparison: For models w/ only one x variable (simple regression), the summation approach is easier and thus preferred. For multi-regression instances, the matrix method is preferred.